Nomenclature.

Basic Structure :

Stereoisomerism ①	Substituents ②	Parent ③	Unsaturation ④	Functional Group ⑤
cis or tranz E or Z R or S	Groups that connected to the main chain (methy, chloro..)	main chain (longest)	double / triple bonds	The group after which the compound is named (oic acid / ol / oate ....)

Ex:-

$\underbrace{(Z)}_{①}-\underbrace{2\text{-chloro}}_{②}\underbrace{\text{pent}}_{③}-\underbrace{2\text{-en}}_{④}-\underbrace{1\text{-ol}}_{⑤}$

Functional Groups.

Carboxylic Acid	$-\overset{O}{\overset{\|\|}{C}}-OH$	-oic acid
Ester	$-\overset{O}{\overset{\|\|}{C}}-OR$	-oate
Amide	$-\overset{O}{\overset{\|\|}{C}}-NH_2$	-amide
Aldehyde	$-\overset{O}{\overset{\|\|}{C}}-H$	-al
Nitrite	$-C\equiv N$	-nitrile
ketone	$-\overset{O}{\overset{\|\|}{C}}-$	-one
Alcohol	$-OH$	-ol
Amine	$-NH_2$	-amine
Ether	$R-O-R'$	-alkoxy
Triple bond	$-C\equiv C-$	-yne
Double bond	$-\underset{\|}{C}=\underset{\|}{C}-$	-ene

Parent Chain.

No of. C atoms	Name
1	- meth
2	- eth
3	- Prop
4	- but
5	- Pent
6	- hex.
7.	- hept
8	- oct
9	- non
10	- dec
11	- undec
12	- dodec
13	- tridec
14	- tetradec
15	- Pentadec

Unsaturation.

Triple Bond

Double Bond

pentane — -ane | pentene — -ene | pentyne — -yne

Substituent Groups.

Esters $-\overset{O}{\overset{\|}{C}}-OR$ -alkoxy

Acyl halides $-\overset{O}{\overset{\|}{C}}-X$ -halofor -myl

Amide $-\overset{O}{\overset{\|}{C}}-NH_2$ -amido

Nitrite $-C\equiv N$ -cyano

Aldehyde $-\overset{O}{\overset{\|}{C}}-H$ -formyl (-oxo)

Ketones $-\overset{O}{\overset{\|}{C}}-$ -oxo

Alcohols $-OH$ -hydroxy

Thiols $-SH$ -mercapto

Amine $-NH_2$ -amino

hydrocarbons -

Ether -oxy

Halides $-X$ halo

(i.) $-F$ Fluoro

(ii.) $-Cl$ chloro

(iii.) $-Br$ Bromo

(iv.) $-I$ Iodo

Nitro compounds $-NO$ nitro

Orgo Compounds.

Aliphatic

- linear structure.

Aromatic

- cyclic structure.

Hydrocarbons. (H & C only)

Saturated

* fully satisfied
* Can't take up more H.
* Carbon - carbon single bonds.

Unsaturated

* Not fully satisfied
* Can take up more H atoms
* Triple and double bonds are present.

Parent Chain could have;

a) Maximum number of Substituents.

b) Maximum number of multiple or double bonds.

c) Maximum length (Maximum number of carbon atoms.)

Alkanes.

① Straight Chain Alkanes.

- only have a first name and last name

\# of carbon atoms + saturation

② With branches.

- Identify parent chain.
- Number it. (lowest # for substituents)
- Name parent & substituents
- May be have to use tri, di ....

③ Alkanes with branched substituents.

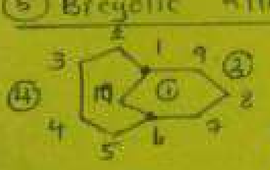
- Here; we have to consider it as a new molecule and name it.

Ex:- (1 - methyl ethyl)

④ cycline alkanes.

- We have to use prefix as 'cyclo' in this case.

⑤ Bicyclic alkanes.



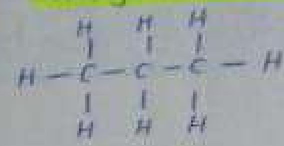
[4,3,1]

use prefix 'bicyclo'

Naming Alkanes.

① Naming straight chain Alkanes.

Ex:-



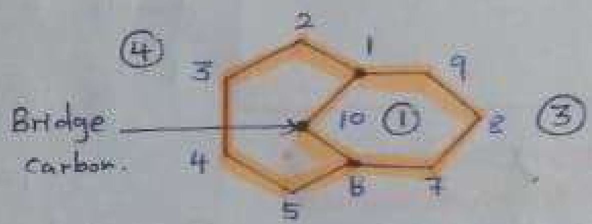
② Naming Branched chain Alkanes.

③ Naming Alkanes with branched substituents.

④ Naming cyclic alkanes.

⑤ Naming bycyclic alkanes.

Bicycloalkanes....



• - bridgehead carbons.

* Number all the carbon atoms after identifying bridgehead carbons.

 - Parent chain.

* Count the # of carbon atoms in two sides of the bridge. Here do not count the bridge-head carbons.

We represent it as [4,3,1] (Highest ⟶ lowest)

Prefix ⟶ bicyclo

Ex:-

bicyclo [4,3,1] decane.

Naming Alkenes.

① chain compounds.
② cyclic compounds.

① Parent chain should be contained
1) Maximum # of C
2) double bond.

② Double bond has priority over substitution when numbering the parent chane.

③ But when the substitution is placed before double bond in priority order; it should be given the lowest number.

Ex —OH & C=C
①   ②

?????

When double bond is in side chain;

$CH_2=C—$ 3

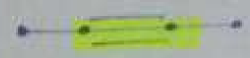
3-ethenyl

$CH_3-CH=CH-CH_2—$ 2

2-butenyl

Naming Alkynes.

• The # of C atoms present in here quiet different.



4-carbon atoms.

3 carbon atoms.

• There are two C atoms in ≡ bond (start & end)

Naming Alkenyue.

(both alkene & alkyne)

* Triple bond has priority over the double bond when numbering.

Ex:—

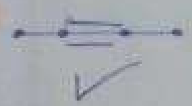
⑤ ④ ③ ② ①
$CH_3—CH=CH—C≡C—H$

3-pentene-1-yne

Special Things about naming Alkynes.

• When there is a terminal C; we give number ① for that.

• Pay attention when have linear structure.

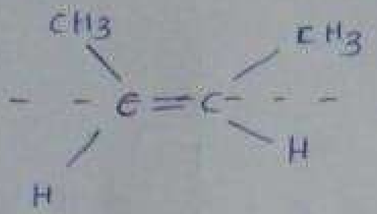


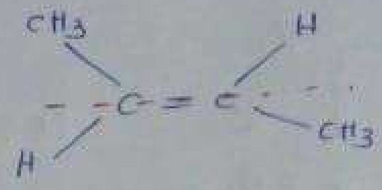
④ carbons
(not ②)

Compounds with other functional groups.

• Things are same in here too.
• Look up the order of substitutional groups and decide the priority.
• Name as same as in Alkyne, Alkene & Alkane.

Cis and Trans Configuration

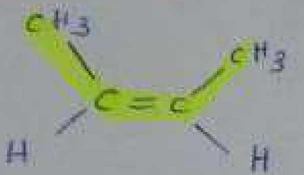




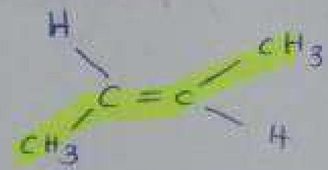
same side → cis

opposit side → trans.

Nomenclature:

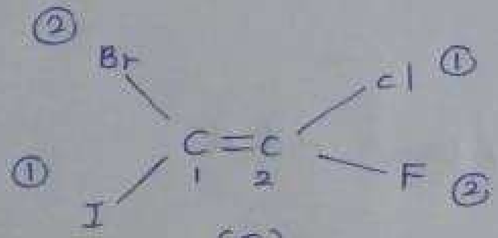


cis-2-butene



trans-2-butere.

E & Z Nomenclature

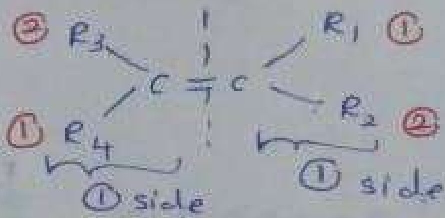


(E)

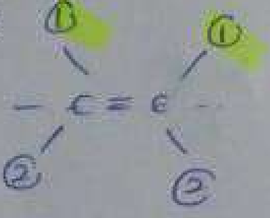
E-1-bromo-2-chloro-2-fluoro-1-iodoethene.

* Here, we consider the atomic number of each atom & give priority numbers 1 & 2.

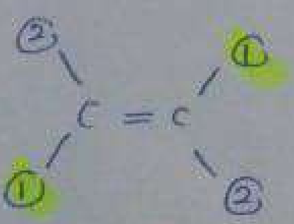
* Here; C atoms are consider like this.



If;

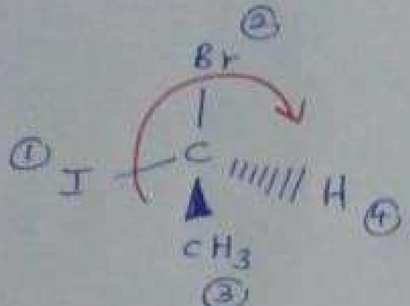


(Z)
(same Z)

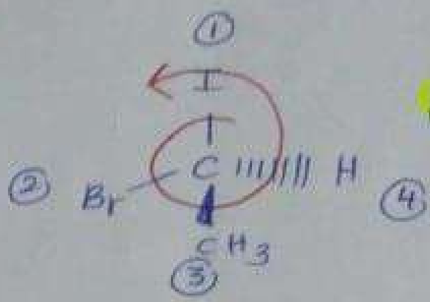


(E)
(opposit)

R & S Nomenclature.



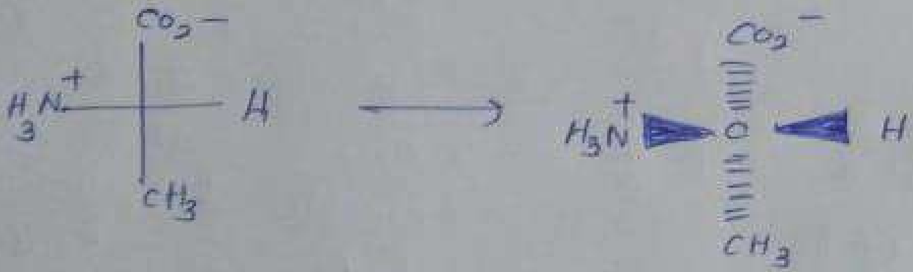
(R)



(S).

Fischer Projection.

If we got fischer structure;



(Think like a bow tie)

Newman Projection